

DEVELOPING A RAPID URBAN FOREST ASSESSMENT SYSTEM FOR
SUSTAINABLE CITY GREENIFICATION

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ABSTRACT

Developing a Rapid Urban Forest Assessment System for Sustainable City Greenification

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This thesis presents the Rapid Urban Forest Assessment (RUFA) system, a web-based platform that integrates urban tree inventories and aerial tree detection to assess forest health across California’s census-designated places. RUFA combines inventoried tree records with coordinates detected from high-resolution multispectral imagery using convolutional neural networks, then computes a composite RUFA Score from four metrics: canopy cover percentage, trees per capita, tree diversity (TD-50), and tree evenness. The thesis addresses two engineering challenges in building the dashboard: querying and aggregating over seven million tree records in real time, and rendering spatial summaries at multiple zoom levels without recomputing cluster assignments on every map interaction. A MySQL schema with indexing and cascading materialized views enables sub-second city-level queries, and a SuperCluster index paired with Voronoi tessellation supports interactive map visualization. Deployed on AWS, RUFA provides urban foresters, planners, and researchers with comparable scores and multi-scale map outputs to support evidence-based urban forestry management.

Keywords: urban forestry, tree density, biodiversity, TD-50, scalable, convolutional neural networks

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Chapter 1

INTRODUCTION

Urban and community forestry (U&CF) programs play a crucial role in nurturing and enhancing urban green spaces, contributing significantly to the environmental, social, and economic well-being of urban communities. To support these initiatives, advanced tools are being developed to evaluate and report on U&CF activities effectively. One such endeavor is the Rapid Urban Forest Assessment (RUFA) system developed at Cal Poly, a web-based platform designed to integrate, analyze, and disseminate comprehensive data on U&CF. This dashboard serves as a centralized hub for urban forestry data collection and analysis, providing actionable insights into the health and management of urban green spaces while seamlessly integrating resources such as those available on selectree.calpoly.edu.

This thesis addresses two engineering problems encountered in building RUFA. The system must query and aggregate over seven million tree records across hundreds of California census-designated places in real time, yet naive storage and retrieval strategies produce multi-second latencies unsuitable for an interactive dashboard. It must also render spatial summaries at multiple zoom levels without recomputing cluster assignments on every pan or zoom event. Meeting these requirements drove the database design choices in chapter 3 and the clustering work in chapter 5.

The specific contributions of this thesis are as follows:

- A MySQL relational schema with indexing and materialized views for sub-second city-level queries over RUFA’s tree inventory, aerial-detection, and geographic boundary data.

- A cascading materialized-view architecture that pre-computes and propagates RUFA Score metrics across census places, ZIP codes, tracts, and counties.
- Evaluation of spatial indexing strategies for assigning tree points to administrative polygons at RUFA scale.
- An AWS-based three-tier architecture for the RUFA dashboard.
- Progressive clustering algorithms, from K-means to SuperClustering with Voronoi diagrams, for multi-scale map visualization.

The rest of this document is organized as follows. chapter 2 establishes the institutional and scientific context for RUFA. The following chapters present the system in depth, covering data storage and query performance, deployment architecture, and interactive map clustering, before chapter 6 summarizes the contributions and discusses future work.

Chapter 2

BACKGROUND

The Rapid Urban Forest Assessment (RUFA) system integrates two primary data sources to provide comprehensive insights into urban forest ecosystems. The first data source offers highly detailed tree characteristics, derived from an extensive compilation of California urban forest inventories. These inventories were aggregated from diverse sources, including private arborist firms (West Coast Arborists, Davey Tree Company, APlus Tree Care) and public entities such as CAL FIRE. This robust dataset ensures a high level of granularity and accuracy in tree attribute information [4]. Our second data source estimates tree coordinates using a deep learning approach. This method employs high-resolution multispectral aerial imagery to detect individual trees in urban settings. A convolutional neural network is utilized to generate a confidence map by regressing the detected tree information, enabling precise spatial quantification of urban tree coverage [10].

2.1 Diversity Metrics

Diversity metrics play a crucial role in understanding the composition and resilience of urban forests. Among these, the TD-50 index, introduced by Love et al. [4], has emerged as a pivotal tool for assessing urban tree species diversity. This index identifies the species that constitute the top 50% of urban forest abundance, providing a snapshot of both ecological dominance and species richness.

The value of such metrics lies in their ability to guide urban forestry programs toward enhanced biodiversity. High diversity levels improve resilience to pests, diseases,

and environmental changes, while low diversity often signifies vulnerability. For instance, McPherson et al. [5] highlight the importance of diversifying tree species as a strategy to bolster urban forest resilience against climate change impacts. Furthermore, Livesley et al. [3] emphasize the ecological functions provided by diverse urban forests, such as improved air quality, habitat provision, and the regulation of urban microclimates.

Research underscores that species diversity influences urban forests' ability to adapt to environmental stressors, including extreme weather events and urban heat islands. Therefore, incorporating diversity metrics into urban forestry management is critical for maintaining ecological stability and sustainability.

2.2 Canopy and Density

Urban tree canopy coverage and density are fundamental indicators of urban forest health and performance. Studies like those by Livesley et al. [3] and Sanusi et al. [8] underscore the significance of these metrics in understanding urban tree contributions to ecological and social well-being.

Canopy coverage is particularly effective in regulating urban temperatures and reducing solar radiation. This is especially relevant for small-statured trees, which Sanusi et al. [8] found to be instrumental in urban cooling, despite their limited size. Additionally, canopy density directly correlates with urban areas' capacity to manage stormwater runoff, as shown in Xiao and McPherson's [11] research.

Density metrics also provide insights into forest sustainability. Higher tree densities often indicate robust forest growth and effective carbon sequestration capabilities. However, overcrowding can lead to competition for resources, reducing overall health

and resilience. Balancing canopy coverage and density is therefore essential for maximizing the benefits of urban forests, from cooling effects and stormwater management to improving urban aesthetics and human well-being.

2.3 RUFA Score

The Rapid Urban Forest Assessment (RUFA) system also provides a composite “RUFA Score” for each census-designated place in California, facilitating an at-a-glance evaluation of urban forest health and sustainability. This score integrates four equally weighted metrics:

- **Canopy cover percentage (CCP)** – Reflects the proportion of land area covered by tree canopy.
- **Trees per capita (TPC)** – Indicates tree distribution relative to population size.
- **Tree diversity (TD)** – Incorporates the TD-50 index [4] to capture species richness and dominance.
- **Tree evenness (TE)** – Describes how uniformly individuals are distributed among available species.

Each metric is *binned* into a score of 1 to 5 based on its distribution across California. Summing these binned values yields a maximum of 20 possible points. The final RUFA Score is then calculated as:

$$\text{RUFA Score} = \left(\frac{\text{CCP} + \text{TD} + \text{TE} + \text{TPC}}{20} \right) \times 100, \quad (2.1)$$

where higher values (closer to 100) indicate stronger overall performance. By normalizing each component and summing their contributions, the RUFA Score offers a standardized, comparable index of urban forest health. This approach ensures that areas with varying population densities, canopy coverage, and species compositions can be evaluated on a consistent scale, guiding strategic planning and policy decisions to enhance urban forest resilience.

Chapter 3

RUFA DATABASE DESIGN

In this chapter, we delve into the rationale behind the relational database for the RUFA project, a system designed to track tree records across various geographic and ecological dimensions. A relational database provides an ideal solution for managing structured data, such as tree attributes, geographic metrics, and hierarchical relationships between places, zip codes, and tracts. We will explore the MySQL workflow, emphasizing data integrity, normalization principles, materialized views for efficient querying, and indexing strategies to optimize database performance. These features ensure scalable, accurate, and high-performance data management for the RUFA project.

3.1 SQL Workflow

A well-defined SQL workflow is crucial for maintaining data integrity and ensuring efficient data retrieval [2]. The following subsections will detail the tables used in the RUFA database, the normalization process adopted, and the overall database schema design.

3.1.1 Tables and Normalization

The RUFA database consists of several tables designed to minimize redundancy and dependency. By normalizing the database to the third normal form (3NF), we ensure

that each table contains data related to a specific topic, thereby enhancing data consistency and simplifying maintenance.

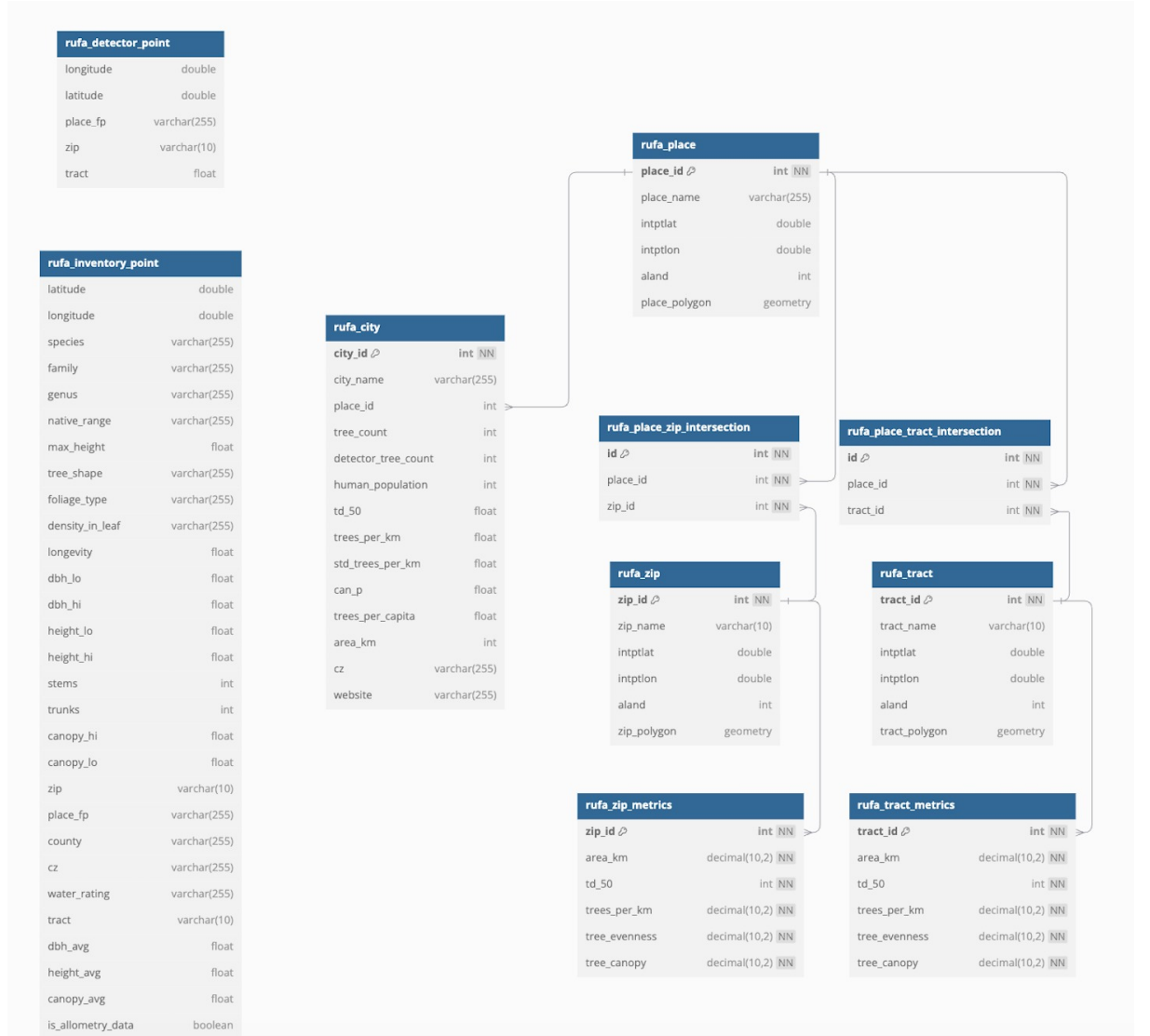


Figure 3.1: Entity Relationship Diagram (ERD) for the RUFA database schema

3.1.2 Entity Relationship Diagram

Figure 3.1 illustrates the ERD for the RUFA database schema. This diagram highlights the relationships between different entities, such as Points, Polygons, and Places, ensuring that the database structure supports all necessary operations and

queries efficiently. Points are derived from our core data sources, while polygons represent geographic boundaries including census places, ZIP codes, and census tracts.

3.1.3 Alternative Database Solutions

While MySQL serves as the database choice for the RUFA, there are several alternative systems that offer enhanced capabilities for spatial data management and analysis. Given that the RUFA project relies on precise geographical information—such as census tracts, city boundaries, and tree coordinates—optimizing spatial queries and performance can be a critical consideration.

PostgreSQL/PostGIS. One of the most commonly cited alternatives for spatially intensive applications is PostgreSQL coupled with its PostGIS extension [7]. PostGIS extends the relational engine to include advanced geospatial data types, indexing methods (e.g., GiST, SP-GiST), and an extensive library of geospatial functions (for instance, `ST_Intersect`, `S_Buffer`, and `ST_Distance`). These features support complex spatial queries, such as determining which trees fall within a specific bounding polygon or calculating nearest-neighbor distances between tree points. Additionally, PostGIS’s robust handling of coordinate reference systems (CRS) and transformations can greatly simplify cross-regional comparisons within the RUFA framework.

MongoDB with Geospatial Indexes. MongoDB, a NoSQL document database, also offers geospatial indexing that can handle queries for distance-based searches on points, lines, and polygons [1]. This schema-free approach can be advantageous for storing heterogeneous tree data—especially when the attributes of different species vary significantly—since new fields can be added without redefining a rigid schema.

GIS-Specific Databases. Some organizations adopt specialized spatial database solutions, such as *Esri's ArcGIS Enterprise* or *Oracle Spatial and Graph*, for high-volume or enterprise-level spatial data management. These systems offer robust toolsets for spatial analyses, but they can come with higher licensing costs and steeper learning curves. For smaller projects or open-source-driven initiatives, this may pose resource constraints and limit broad community involvement.

Why MySQL Remains in Use. Despite these more spatially advanced alternatives, MySQL remains the primary choice for the RUFA project due to its historical ties with the selectree.calpoly.edu infrastructure. Many existing scripts, tools, and user-facing applications are already deeply integrated with MySQL, making a complete or partial transition to another platform expensive in terms of development effort and time. Moreover, while MySQL's native spatial features are comparatively limited, they can still manage a variety of spatial queries by leveraging *SPATIAL indexes* on MyISAM or InnoDB tables.

For MyISAM and InnoDB tables, search operations in columns containing spatial data can be optimized using SPATIAL indexes. The most typical operations are:

- *Point queries* that search for all objects containing a given point
- *Region queries* that search for all objects overlapping a specified region

MySQL uses *R-Trees* with quadratic splitting for SPATIAL indexes on geometry columns. A SPATIAL index is built using the *minimum bounding rectangle* (MBR) of a geometry. For most geometries, the MBR is the smallest rectangle that completely encloses the shape. For a horizontal or vertical linestring, the MBR degenerates to that linestring; and for a point, it degenerates to a single point. It is also possible

to create normal (non-SPATIAL) indexes on spatial columns, although these indexes have more restrictions—for instance, in a non-SPATIAL index, you must generally restrict the data to a specific storage engine or data type to ensure efficient lookups.

Ultimately, although PostgreSQL/PostGIS/MongoDB with geospatial indexes, or specialized GIS databases may outperform MySQL in certain spatial workflows, the RUFA project’s current operational environment and alignment with Selectree’s architecture underscore MySQL’s continued use. Future iterations of RUFA could incorporate or migrate to these alternatives if spatial demands grow significantly—potentially leveraging hybrid architectures that pair MySQL’s relational strengths with specialized spatial extensions in a multi-database ecosystem.

3.2 Materialized Views

Materialized views are used in the RUFA database to store the results of complex queries, thereby improving query performance. To ensure that these views remain up-to-date with the underlying data, we implement automatic refresh mechanisms. This can be achieved using database triggers or scheduled jobs that periodically update the materialized views based on changes in the base tables.

3.3 Indexing Strategies

Effective indexing is essential for optimizing query performance in the RUFA database. By carefully selecting which columns to index, we can significantly reduce query execution time. Automatic updates to indexes are managed by the database system, which maintains the indexes as data is inserted, updated, or deleted. Additionally,

regular index maintenance tasks, such as rebuilding fragmented indexes, are scheduled to ensure optimal performance.

3.3.1 Automatic Index Maintenance

Modern relational database management systems (RDBMS) handle index maintenance automatically. However, for large databases like RUFA, it’s advisable to monitor index performance and perform manual maintenance when necessary. Tools and scripts can be employed to automate these maintenance tasks, ensuring that indexes remain efficient without manual intervention.

3.4 Materialized View Implementation for RUFA Score Management

3.4.1 Overview

The RUFA system relies on complex calculations involving multiple metrics—canopy cover percentage (CCP), trees per capita (TPC), tree diversity (TD-50), and tree evenness (TE)—to generate composite RUFA scores for census-designated places across California. Given the computational intensity of these calculations and the need for responsive user queries, implementing materialized views becomes essential for maintaining system performance while ensuring data accuracy.

The system faces a unique challenge in that geographic areas are hierarchically related—when tree data changes in a city like Los Angeles, the system must update not only the city-level statistics but also all associated ZIP codes, census tracts, climate zones, and county-level aggregations. This cascading update requirement necessitates a sophisticated materialized view strategy that can efficiently propagate changes through the geographic hierarchy.

3.4.2 Geographic Cascade Architecture

Since MariaDB does not natively support true materialized views like PostgreSQL, the RUFA system implements a specialized architecture using regular tables that function as materialized views, combined with automated refresh mechanisms to maintain data consistency across geographic hierarchies.

The core innovation lies in the cascading update system that recognizes the interconnected nature of geographic boundaries. When tree inventory data changes in any location, the system automatically identifies and updates all related geographic areas through a relationship mapping table and stored procedure framework. This ensures that statistics remain synchronized across ZIP codes, cities, census tracts, climate zones, and counties without requiring manual intervention.

3.4.2.1 Unified Statistics Table

The system maintains a central `tree_stats_by_location` table that stores pre-calculated metrics for all geographic location types within a single, normalized structure. This table contains comprehensive tree statistics including species diversity counts, average tree characteristics, native species distributions, and water requirement breakdowns. Each record is identified by a location type (zip, city, tract) and corresponding location code, enabling efficient querying across different geographic scales.

3.4.2.2 Relationship Mapping and Cascade Logic

A separate `location_relationships` table maintains the hierarchical connections between different geographic boundaries, enabling the system to determine which areas need updates when tree data changes. The cascade logic is implemented through

stored procedures that automatically refresh statistics for all related geographic areas when triggered by data modifications.

For example, when tree inventory data changes in Los Angeles, the system automatically updates statistics for the city itself, all ZIP codes within Los Angeles boundaries, associated census tracts, the relevant climate zone, and Los Angeles County. This ensures consistency across all geographic scales without requiring users or administrators to manually trigger updates for each affected area.

Trigger-Based Updates Database triggers are implemented on core tables to automatically invalidate and schedule refreshes of dependent materialized views:

***Code Listing 3.1:** Database trigger for automatic refresh scheduling*

```
DELIMITER $
CREATE TRIGGER update_rufa_scores_trigger
AFTER INSERT ON tree_inventory
FOR EACH ROW
BEGIN
    UPDATE mv_refresh_schedule
    SET needs_refresh = TRUE,
        last_modified = NOW()
    WHERE view_name = 'mv_rufa_scores'
    AND place_id = NEW.place_id;
END$
DELIMITER ;
```

Scheduled Batch Processing A scheduled job runs every N hours to process bulk updates and recalculate scores for modified areas:

***Code Listing 3.2:** Batch refresh procedure for RUFA scores*

```
-- Batch refresh procedure
DELIMITER $
CREATE PROCEDURE RefreshRUFAScores()
BEGIN
    DECLARE done INT DEFAULT FALSE;
```



```

DECLARE place_to_update VARCHAR(50);

-- Cursor for places needing updates
DECLARE place_cursor CURSOR FOR
    SELECT DISTINCT place_id
    FROM mv_refresh_schedule
    WHERE needs_refresh = TRUE;

DECLARE CONTINUE HANDLER FOR NOT FOUND SET done =
    TRUE;

OPEN place_cursor;

refresh_loop: LOOP
    FETCH place_cursor INTO place_to_update;
    IF done THEN
        LEAVE refresh_loop;
    END IF;

    -- Recalculate RUFA score components
    CALL CalculateRUFAScore(place_to_update);

    -- Mark as refreshed
    UPDATE mv_refresh_schedule
    SET needs_refresh = FALSE,
        last_refreshed = NOW()
    WHERE place_id = place_to_update;

END LOOP;

CLOSE place_cursor;
END$
DELIMITER ;

```

Incremental Update Strategy Rather than full table refreshes, the system implements incremental updates that only recalculate scores for affected geographic areas:

- **Spatial Partitioning:** RUFA scores are partitioned by climate zones and counties to isolate updates

- **Change Detection:** Modified tree records trigger updates only for their specific census-designated places
- **Dependency Tracking:** A dependency graph ensures that changes to base data propagate correctly through all dependent materialized views

3.4.3 Data Consistency Safeguards

To maintain data integrity during the refresh process:

3.4.3.1 Version Control

Each materialized view maintains version numbers to detect stale data:

***Code Listing 3.3:** Version control table for materialized views*

```
CREATE TABLE mv_version_control (
    view_name VARCHAR(50) PRIMARY KEY,
    current_version INT NOT NULL DEFAULT 1,
    last_refresh TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    refresh_duration_seconds INT,
    status ENUM('CURRENT', 'REFRESHING', 'STALE')
    DEFAULT 'CURRENT'
);
```

3.4.3.2 Atomic Updates

Large refresh operations use temporary tables and atomic swaps to prevent serving inconsistent data:

***Code Listing 3.4:** Atomic table swap for consistent updates*

```
-- Create temporary table with updated scores
CREATE TABLE mv_rufa_scores_temp LIKE mv_rufa_scores;
```

```

-- Populate with fresh calculations
INSERT INTO mv_rufa_scores_temp
SELECT place_id,
       ((ccp_score + td_score + te_score + tpc_score) /
        20) * 100 as rufa_score,
       NOW() as calculated_at
FROM mv_tree_metrics
JOIN mv_population_data USING (place_id);

-- Atomic swap
RENAME TABLE mv_rufa_scores TO mv_rufa_scores_old,
              mv_rufa_scores_temp TO mv_rufa_scores;

DROP TABLE mv_rufa_scores_old;

```

3.4.4 Performance Optimization

The materialized view system includes several performance optimizations:

3.4.4.1 Selective Refresh

Only geographic areas with actual data changes are recalculated, dramatically reducing refresh times from hours to minutes for typical updates.

3.4.4.2 Parallel Processing

Multiple refresh workers can simultaneously update different geographic partitions, leveraging MariaDB's parallel processing capabilities.

3.4.4.3 Caching Strategy

Frequently accessed RUFA scores are cached in Redis with appropriate TTL values, reducing database load for dashboard queries.

3.4.5 Monitoring and Alerting

The system includes comprehensive monitoring to ensure materialized views remain current:

- **Staleness Detection:** Automated alerts when materialized views fall behind source data by more than 12 hours
- **Performance Monitoring:** Tracking of refresh times and resource utilization
- **Data Quality Checks:** Validation that recalculated scores fall within expected ranges

This materialized view architecture ensures that RUFA scores remain accurate and current while providing the query performance necessary for responsive user interactions with the Urban Forestry Assessment Dashboard.

3.4.6 Indexing Results

To evaluate the performance impact of different data storage and indexing strategies, we conducted comprehensive benchmarking tests across various query scenarios. Our analysis compared four different approaches for data retrieval and rendering within the RUFA system, measuring query execution times and system responsiveness under typical usage patterns. All performance tests were conducted on a standardized test

environment consisting of an AWS EC2 instance (t3.large) with 2 vCPUs, 8GB RAM, and 10.5.23 MariaDB running on Amazon RDS. The test dataset contained approximately 7.2 million tree records distributed across 702 cities in California, matching the production data scale described in the California Urban Forest Inventory.

Each test scenario was executed 100 times to ensure statistical significance, with query execution times measured using MySQL’s built-in profiling tools. The performance metrics were calculated as:

$$\text{Throughput} = \frac{1}{\text{Average Query Time (seconds)}} \quad (3.1)$$

$$\text{Performance Improvement} = \frac{T_{\text{baseline}} - T_{\text{optimized}}}{T_{\text{baseline}}} \times 100\% \quad (3.2)$$

where T_{baseline} represents the baseline query time and $T_{\text{optimized}}$ represents the optimized query time.

3.4.7 Performance Comparison Results

Table 3.1 presents the average query execution times across different storage and indexing strategies for city-specific queries retrieving tree inventory data.

Table 3.1: *Performance comparison of different storage and indexing strategies*

Storage Method	Avg. Time (ms)	Std. Dev.	Queries/s
S3 Single CSV File	8,450	1,230	0.12
S3 Per-City CSV Files	3,120	480	0.32
Database Table (No Index)	2,890	290	0.35
Database Table (Index)	145	25	6.90

3.4.7.1 S3 Single CSV File Approach

The initial approach of storing all tree data in a single CSV file on Amazon S3 demonstrated the poorest performance, with average query times exceeding 8.4 seconds. This method required downloading and parsing the entire dataset for each query, regardless of the specific city or region being requested. The high standard deviation ($\sigma = 1,230$ ms) indicated inconsistent performance, likely due to network latency variations and the overhead of processing large file transfers.

3.4.7.2 S3 Per-City CSV Files

Partitioning the data into individual CSV files per city showed significant improvement, reducing average query times to 3.12 seconds—a 63% performance improvement over the single file approach. This approach eliminated the need to process irrelevant data for city-specific queries. However, the system still faced overhead from file system operations and CSV parsing, limiting its scalability for real-time applications.

3.4.7.3 Database Table Without Indexing

Direct queries against the MySQL database table without proper indexing yielded slightly better performance than the partitioned CSV approach, with average query times of 2.89 seconds. While this eliminated file parsing overhead, the lack of indexing on frequently queried columns (particularly city names and geographic identifiers) required full table scans with $O(n)$ complexity for most operations.

3.4.7.4 Database Table With City Indexing

The implementation of a B-tree index on the city column resulted in dramatic performance improvements, reducing average query times to just 145 milliseconds—a 95% improvement over the non-indexed approach. The low standard deviation ($\sigma = 25$ ms) demonstrates consistent performance across different query patterns. This approach achieved a throughput of 6.90 queries per second, making it suitable for interactive web applications and real-time dashboard updates.

3.4.8 Memory and Storage Impact

The indexing strategy increased database storage requirements by approximately 15%, from 2.3GB to 2.6GB for the complete dataset. However, this storage overhead was offset by significant reductions in memory usage during query execution, as indexed queries required substantially less working memory for result set processing.

The storage overhead can be expressed as:

$$\text{Storage Overhead} = \frac{S_{\text{indexed}} - S_{\text{original}}}{S_{\text{original}}} \times 100\% = \frac{2.6 - 2.3}{2.3} \times 100\% = 13.0\% \quad (3.3)$$

3.4.9 Scaling Considerations

Performance testing with varying dataset sizes demonstrated that the indexed approach maintains logarithmic time complexity $O(\log n)$, ensuring scalability as the urban forest inventory continues to grow. Projections suggest that even with a 10× increase in data volume, indexed query times would remain under 300ms for typical

city-based queries, maintaining the system's responsiveness for interactive applications.

Chapter 4

SYSTEM DESIGN

The rapid urban forest assessment system requires a robust and scalable architecture capable of handling large-scale tree inventory data while providing real-time query capabilities for municipal planners and environmental researchers. Building upon the performance analysis presented in Chapter 3, which demonstrated that indexed database queries achieve 95% performance improvements over non-indexed approaches, this chapter outlines the comprehensive system design that enables efficient processing and visualization of urban forest data across California’s 702+ cities.

The system architecture addresses three critical design requirements: scalability to accommodate the 7.2 million tree records in the California Urban Forest Inventory, performance optimization to support interactive web applications with sub-second response times, and extensibility to incorporate additional environmental datasets and analysis capabilities. The design leverages cloud-native technologies and modern web frameworks to create a maintainable and cost-effective solution.

This chapter is organized into three main sections that detail the key architectural components. Section 4.1 presents the infrastructure utilized, including a comprehensive system diagram, the optimized database design that achieves 6.90 queries per second throughput, S3 integration with ESRI connectivity for geospatial data processing, and scalability considerations for future growth. Section 4.2 outlines the security and permissions framework, covering user store management, role-based permissions architecture, and NGINX configuration for secure request handling. Finally, Section 4.3 describes the deployment strategy, focusing on the continuous integration

and continuous deployment (CI/CD) pipeline that ensures reliable system updates and maintenance.

The resulting system design provides a foundation for sustainable city greenification efforts by enabling rapid assessment of urban forest conditions, identification of tree planting opportunities, and data-driven decision making for municipal environmental programs.

4.1 Infrastructure Used

4.1.1 System Diagram

4.1.2 Database

4.1.3 S3 - ESRI Connectivity

4.1.4 Scalability

4.2 Security and Permissions

4.2.1 User Store

4.2.2 Permissions

4.2.3 NGINX

4.3 Deployment

4.3.1 CI/CD

Chapter 5

CLUSTERING

The clustering component of RUFA groups millions of tree coordinates for interactive map visualization at multiple zoom levels. This chapter describes the algorithms evaluated during development, from an initial K-means prototype through Super-Clustering with Voronoi tessellation, and the performance trade-offs that motivated each transition.

5.1 K-Means

The first implementation applied Lloyd’s K-means algorithm to tree coordinates, minimizing the within-cluster sum of squares (WCSS):

$$\text{WCSS} = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2, \quad (5.1)$$

where C_i is cluster i , μ_i its centroid, and x an individual tree location. Centroids were initialized with K-means++, distances were computed with the Haversine formula, and iteration continued until centroid movement fell below 0.001° or 100 iterations were reached.

Because cluster count must be specified in advance, k was chosen dynamically as a function of the number of input points and the map zoom level, typically ranging from 50 to 500 clusters. The intent was to preserve local structure in dense areas while limiting marker overlap at wider views. On Los Angeles data, scaling k with

input size still produced overlapping clusters in dense neighborhoods and failed to yield stable multi-scale summaries when the viewport changed.

Even when k increased with point count, clusters in high-density neighborhoods remained visually indistinguishable, and recomputing assignments for each zoom or pan interaction incurred $O(nkt)$ cost that did not scale to California’s full inventory. Benchmarking on 100,000 trees averaged 2.3 seconds per run with 45 MB of working memory. K-means provided a workable baseline but could not meet RUFA’s requirement for responsive, zoom-aware aggregation over millions of points.

5.2 Intermediate Approaches

Several refinements were explored before adopting a hierarchical tile-based strategy. Result caching stored centroid positions and cluster assignments across repeated queries, reducing redundant computation when users returned to previously viewed regions, but did not address the fundamental cost of recomputing clusters at each zoom level.

5.3 SuperClustering with Voronoi Diagrams

The production implementation combines SuperCluster, a hierarchical point-clustering library used in web mapping, with Voronoi tessellation to define spatial analysis regions at each zoom level. SuperCluster precomputes a pyramid of clusters from zoom 0 to 10; at query time, RUFA requests only the clusters intersecting the current bounding box, avoiding full-dataset recomputation on pan and zoom events.

Code Listing 5.1: *SuperCluster initialization and zoom-level processing*

```
index = pysupercluster.SuperCluster(  
    detector_points,
```

```

        min_zoom=0,
        max_zoom=10,
        radius=5,
        extent=512
    )

    for zoom in range(5, 10):
        clusters = index.getClusters(bbox, zoom)
        if len(clusters) >= 4:
            voronoi_data = process_voronoi(clusters, zoom)

```

For zoom levels with at least four cluster centroids, RUFA generates a Voronoi diagram from the centroid set and converts each cell into a polygon bounded by the viewport. Trees are assigned to cells by point-in-polygon containment, and each cell receives tree density (trees per km²) and a TD-50 diversity score computed from species counts within the cell. Invalid or unbounded Voronoi regions are clipped to the query extent before export as GeoJSON features for the dashboard.

SuperCluster initialization runs in $O(n \log n)$ time; Fortune’s algorithm generates the Voronoi diagram in $O(m \log m)$ for m centroids, and point assignment is $O(mn)$ in the naive formulation used during development. In practice, the hierarchical index limits m to the clusters visible at each zoom, and cached tile results make zoom-level switching near instantaneous. Higher zoom levels expose local density variation; lower zoom levels collapse detail into regional summaries suitable for statewide comparison. Applied to San Luis Obispo, the final pipeline partitions the city into Voronoi cells aligned with SuperCluster centroids, with color mapping reflecting per-cell tree density. This replaces the overlapping K-means markers with a stable, zoom-aware representation suitable for statewide deployment.

Chapter 6

CONCLUSIONS

This thesis addressed two engineering problems encountered in building the Rapid Urban Forest Assessment (RUFA) dashboard: querying and aggregating millions of tree records across hundreds of California census-designated places in real time, and producing stable spatial summaries at multiple zoom levels without recomputing cluster assignments on every map interaction. Each problem was approached by characterizing the workload, prototyping a straightforward solution, measuring where it broke down, and replacing it with an implementation that scales to the production dataset.

On the data side, RUFA’s MySQL schema combines normalized inventory and aerial-detection tables with cascading materialized views that propagate RUFA Score metrics from census places to ZIP codes, tracts, and counties. Spatial indexing strategies were evaluated for assigning tree points to administrative polygons, and indexed queries achieved sub-second response times against the full California inventory. The resulting design preserves compatibility with the existing `selectree.calpoly.edu` infrastructure while delivering the interactive performance required by the dashboard.

On the visualization side, the clustering pipeline evolved from a fixed- k K-means baseline, which produced overlapping markers in dense neighborhoods and recomputed assignments on every zoom event, to a SuperCluster index paired with Voronoi tessellation. The final implementation precomputes a multi-resolution cluster hierarchy and derives per-cell tree density and TD-50 diversity scores at query time, yielding zoom-aware summaries that match the uneven density of urban tree distributions.

Together with the AWS-based three-tier deployment described in chapter 4, these contributions form a system that aggregates inventory and aerial-detection data, computes the four-component RUFA Score, and serves interactive assessments to urban foresters, planners, and researchers. The composite score reduces a complex, multi-source dataset to a single comparable index, supporting evidence-based decisions about planting, maintenance, and policy across California communities.

6.1 Future Work

Several directions extend RUFA beyond the scope of this thesis.

6.1.1 Allometric Estimation

UFEI’s AllomeTree tool [9] exposes species-specific allometric equations for California’s urban forest, predicting diameter at breast height (DBH), height, and canopy width with 95% prediction intervals from either DBH or tree age. The equations derive from the U.S. Forest Service Urban Tree Database [6], which fits 365 equation sets across 171 species and 16 climate regions. Integrating these models into RUFA would let the system estimate structural attributes for trees with incomplete inventory records and for the aerial-only detections produced by the Ventura et al. pipeline, which currently provide location and canopy extent but no DBH or height. With those estimates, RUFA could extend its reporting to ecosystem services such as carbon storage, leaf area, and stormwater interception, derived per-tree rather than aggregated from sparse inventory measurements.

6.1.2 Incremental Data Integration

The current materialized-view refresh runs as a batch process. Incremental refresh triggered by inventory updates or new aerial detections would keep RUFA Scores current without recomputing every dependent view, and would support partner organizations contributing inventory data on a rolling basis.

6.1.3 Mobile and Field Interfaces

A field-oriented interface would let urban foresters capture or correct inventory records on site and view RUFA assessments from mobile devices. Bidirectional sync between field captures and the central database would tighten the loop between assessment and on-the-ground management.

6.1.4 Geographic Expansion

Extending RUFA beyond California requires adapting the schema, score normalization, and species references to other climate zones and regulatory contexts. The data model and clustering pipeline are state-agnostic; the main work lies in re-binning each metric against a new statewide distribution and incorporating regional species lists.

As urban areas continue to grow and face increasing environmental pressure, tools that quantify urban forest structure, diversity, and equity will become more important to municipal planning. RUFA contributes to that toolset by combining automated detection, inventory aggregation, and interactive visualization in a single, comparable assessment.

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